Phase-adjusted realification of a $\mathbb{C}^3$ Kochen-Specker configuration into $\mathbb{R}^6$

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We describe a phase-adjusted realification procedure that embeds any finite set of rays in $\mathbb{C}^3$ into $\mathbb{R}^6$. By assigning an appropriate phase to each ray before applying the standard coordinate-wise map, we can arrange that two rays are orthogonal in $\mathbb{C}^3$ if and only if their images are orthogonal in $\mathbb{R}^6$, so the construction yields a faithful orthogonal representation of the original complex configuration. As a concrete example, we consider the 165 projectively distinct rays used in a $\mathbb{C}^3$ Kochen–Specker configuration obtained from mutually unbiased bases, list these 165 rays explicitly in $\mathbb{C}^3$, and give for each of them its image in $\mathbb{R}^6$ under the canonical realification map. We also note that, because the original 3-element contexts are no longer maximal in $\mathbb{R}^6$, the embedded configuration admits two-valued states even though its realisation with maximal contexts in $\mathbb{C}^3$ is Kochen–Specker uncolourable.

I. A KOCHEN-SPECKER PROOF FOR THE NONEQUIVALENCE OF REAL AND COMPLEX HILBERT SPACE

A foundational question in quantum theory is whether the use of complex numbers is merely a matter of mathematical convenience or a fundamental feature of the physical formalism [1]. On the one hand, any finite-dimensional complex Hilbert space $\mathbb{C}^n$ can be represented as a real Hilbert space $\mathbb{R}^{2n}$ via the usual "realification" map that separates real and imaginary parts. From this perspective, it is tempting to regard real and complex quantum theories as essentially equivalent. On the other hand, there is increasing evidence that the structure of complex Hilbert spaces permits logical and physical phenomena that cannot be reproduced if one insists on a purely real description [2–4].

In this paper we present a complementary, fully deterministic argument based on the logical framework of the Kochen–Specker (KS) theorem. Rather than working with inequalities and probabilities, the KS approach works with a finite set of rays and their orthogonality relations and derives a direct contradiction with the existence of classical truth assignments formalized by two-valued states. We focus on a specific KS-type configuration introduced by Cabello [5] as a threefold extension of the Yu–Oh configuration [6].

As had been earlier pointed out by Harding and Salinas Schmeis [7] and analysed in detail by Navara and Svozil [8], a key structural ingredient of this configuration is the existence, in $\mathbb{C}^3$, of a complete set of mutually unbiased bases (MUBs). Two orthonormal bases are mutually unbiased if every vector of one basis has equal "overlap" with every vector of the other: whenever the system is prepared in a state from one basis, a measurement in the other basis yields each of its D outcomes with equal probability $1/D$. Measurements in one basis thus randomise the outcomes of measurements in the other. The theory of MUBs is a rich subject with deep connections to quantum information and finite geometry [9–14]. For the present work, the crucial facts are:

1. In $\mathbb{C}^3$ one can construct a complete set of $D+1=4$ MUBs.
2. In $\mathbb{R}^3$ it is impossible to find even two mutually unbiased orthonormal bases.

The Yu-Oh-Cabello-KS configuration we consider [5] is built explicitly from the four MUBs in $\mathbb{C}^3$; its 165 rays and 130 contexts encode an orthogonality hypergraph that has no realisation in $\mathbb{R}^3$.

Our contribution is twofold. First, we recast the Harding–Salinas Schmeis result in a MUB-based setting by exhibiting an explicit 165-ray, 130-context KS configuration in $\mathbb{C}^3$ whose orthogonality relations cannot be realised by any set of unit vectors in $\mathbb{R}^3$. This provides a concrete finite example of a logical structure—a set of propositions and contexts—that exists in three-dimensional complex quantum theory but is impossible in three-dimensional real quantum theory.

Second, we show that the same orthogonality hypergraph does admit a faithful orthogonal representation [15] in $\mathbb{R}^6$: by appropriately choosing the phase of each ray before applying the standard coordinate-wise separation into real and imaginary parts, we embed all 165 rays as vectors in $\mathbb{R}^6$ so that two rays are orthogonal in $\mathbb{C}^3$ if and only if their images are orthogonal in $\mathbb{R}^6$. However, in this real embedding each 3-element context now spans only a 3-dimensional subspace of $\mathbb{R}^6$ and is no longer maximal. It can always be extended to a full orthonormal basis of $\mathbb{R}^6$ by adding three extra vectors outside the configuration. This allows one to construct two-valued states by assigning value 0 to all 165 embedded rays and value 1 to a suitable extra basis vector in each completed context.

In this way, the same finite orthogonality hypergraph, realised as maximal contexts in $\mathbb{C}^3$, is KS-uncolourable

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and admits no classical two-valued states, whereas its faithful realisation in $\mathbb{R}^6$ does admit classical two-valued states once contexts are interpreted as maximal in the larger space. This provides a deterministic, non-probabilistic witness of the logical inequivalence between three-dimensional complex and real quantum theory and complements earlier correlation-based separations between real and complex Hilbert-space models.

II. PHASE-ADJUSTED REALIFICATION IN $\mathbb{C}^3$

A. Set-up

Let $\{\,[\psi_k]\,\}_{k=1}^N$ be a finite set of rays in $\mathbb{C}^3$, with normalized representatives $\psi_k \in \mathbb{C}^3$, and let

$$c_{k\ell} := \langle \psi_k, \psi_\ell \rangle_{\mathbb{C}} \tag{1}$$

denote their complex inner products.

For each ray we may multiply ψ_k by an arbitrary unit complex phase $e^{\mathrm{i}\theta_k}$ without changing the ray. We exploit this freedom and define a *phase-adjusted realification*

$$R_k := \Phi_0\big(e^{\mathrm{i}\theta_k}\,\psi_k\big) \in \mathbb{R}^6, \tag{2}$$

where $\Phi_0 : \mathbb{C}^3 \to \mathbb{R}^6$ with

$$(z_1, z_2, z_3) \mapsto (\Re z_1, \Re z_2, \Re z_3, \Im z_1, \Im z_2, \Im z_3). \tag{3}$$

Since the real part of the complex inner product is exactly the real dot product,

$$\begin{aligned} R_k \cdot R_\ell &= \Phi_0(e^{\mathrm{i}\theta_k}\psi_k) \cdot \Phi_0(e^{\mathrm{i}\theta_\ell}\psi_\ell) \\ &= \Re\Big\langle e^{\mathrm{i}\theta_k}\psi_k, e^{\mathrm{i}\theta_\ell}\psi_\ell \Big\rangle_{\mathbb{C}} \\ &= \Re\big(e^{\mathrm{i}(\theta_\ell-\theta_k)} c_{k\ell}\big). \end{aligned} \tag{4}$$

If $c_{k\ell} = 0$, then $R_k \cdot R_\ell = 0$ for all phases, so complex orthogonality is always preserved. The only potential issue is that for $c_{k\ell} \neq 0$ we might accidentally choose phases such that $R_k \cdot R_\ell = 0$, thereby introducing a spurious orthogonality.

B. Forbidden phase differences for non-orthogonal pairs

Assume $c_{k\ell} \neq 0$. Write $c_{k\ell} = |c_{k\ell}| e^{\mathrm{i}\varphi_{k\ell}}$. Then Eq. (4) becomes

$$R_k \cdot R_\ell = |c_{k\ell}|\,\cos\big((\theta_\ell - \theta_k) + \varphi_{k\ell}\big). \tag{5}$$

Thus $R_k \cdot R_\ell = 0$ if and only if

$$(\theta_\ell - \theta_k) + \varphi_{k\ell} \equiv \frac{\pi}{2} \pmod{\pi}. \tag{6}$$

For a fixed θ_k and fixed nonzero $c_{k\ell}$, this excludes at most two values of θ_ℓ modulo 2π:

$$\theta_\ell \equiv -\varphi_{k\ell} + \frac{\pi}{2} \quad \text{or} \quad \theta_\ell \equiv -\varphi_{k\ell} + \frac{3\pi}{2} \pmod{2\pi}. \tag{7}$$

C. Existence of a faithful embedding

We construct the phases $\theta_1, \dots, \theta_N$ inductively.

- Set $\theta_1 := 0$ arbitrarily.
- Suppose $\theta_1, \dots, \theta_{m-1}$ have been chosen such that for all non-orthogonal pairs (k, ℓ) with $1 \le k < \ell < m$ we have $R_k \cdot R_\ell \neq 0$.
- For the new ray m, consider all pairs (k, m) with $1 \le k < m$ and $c_{km} \neq 0$. For each such pair, the two forbidden values of θ_m described above define a finite set $F_{k,m} \subset S^1$.

Let

$$F_m := \bigcup_{\substack{1\le k<m \\ c_{km}\neq 0}} F_{k,m}, \tag{8}$$

which is a finite subset of the unit circle. Choose θ_m anywhere on $S^1 \setminus F_m$. By construction, for every non-orthogonal pair (k, m) we have $R_k \cdot R_m \neq 0$, while pairs with $c_{km} = 0$ remain orthogonal for all phases. Since the forbidden set F_m is finite and the circle is connected, such a θ_m always exists; in fact, all but finitely many phases work. Induction in m shows that we obtain a full set of phases $\{\theta_k\}_{k=1}^N$ such that

$$c_{k\ell} = 0 \iff R_k \cdot R_\ell = 0. \tag{9}$$

Thus $\{R_k\}_{k=1}^N \subset \mathbb{R}^6$ is a faithful orthogonal representation of the original finite set of rays in $\mathbb{C}^3$.

In particular, if $\{\psi_i, \psi_j, \psi_k\}$ form an orthonormal basis of $\mathbb{C}^3$, then each pair is orthogonal in $\mathbb{C}^3$, hence each pair R_i, R_j, R_k is orthogonal in $\mathbb{R}^6$, so orthogonal triples (contexts) are preserved.

III. EXPLICIT LIST OF THE 165 RAYS AND THEIR IMAGES IN $\mathbb{R}^6$

A. Complex notation

We use the primitive third root of unity

$$\omega := e^{2\pi\mathrm{i}/3} = -\frac{1}{2} + \mathrm{i}\,\frac{\sqrt{3}}{2}, \qquad \omega^2 = \overline{\omega} = -\frac{1}{2} - \mathrm{i}\,\frac{\sqrt{3}}{2}, \tag{10}$$

with $1 + \omega + \omega^2 = 0$.

Every coordinate in the 165 rays below is a real multiple of 1, ω, or ω^2 with integer coefficients in $\{-2, -1, 0, 1, 2\}$. For any $z = a + b\,\omega + c\,\omega^2$ ($a, b, c \in \mathbb{R}$) we have

$$\Re z = a - \frac{b+c}{2}, \tag{11}$$

$$\Im z = \frac{\sqrt{3}}{2}\,(b - c). \tag{12}$$

In particular, $0 \mapsto (0,0)$, and

$$\begin{aligned} 1 &\mapsto (1,0), & -1 &\mapsto (-1,0), \\ \omega &\mapsto \left(-\tfrac{1}{2},\ \tfrac{\sqrt{3}}{2}\right), & \omega^2 &\mapsto \left(-\tfrac{1}{2},\ -\tfrac{\sqrt{3}}{2}\right), \\ 2\omega &\mapsto \left(-1,\ \sqrt{3}\right), & 2\omega^2 &\mapsto \left(-1,\ -\sqrt{3}\right), \\ -\omega &\mapsto \left(\tfrac{1}{2},\ -\tfrac{\sqrt{3}}{2}\right), & -\omega^2 &\mapsto \left(\tfrac{1}{2},\ \tfrac{\sqrt{3}}{2}\right), \\ -2\omega &\mapsto \left(1,\ -\sqrt{3}\right), & -2\omega^2 &\mapsto \left(1,\ \sqrt{3}\right). \end{aligned} \tag{13}$$

Since three complex components are encoded as six real ones, the resulting $\mathbb{R}^6$ representation is not unique: any permutation of the six real coordinates gives an orthogonally equivalent configuration. In particular, there are $3! \cdot 2^3 = 48$ permutations that merely reshuffle the three complex entries and, for each entry, possibly swap its real and imaginary part. For example, the vector $(1,1,1,\ 0,0,0)$ is permutationally equivalent to $(1,0,1,0,1,0)$ via the permutation $(2\,4\,5\,3)$ on $\{1,\ldots,6\}$; that is,

$$(x_1,x_2,x_3,x_4,x_5,x_6) \mapsto (x_1,x_4,x_2,x_5,x_3,x_6).$$

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B. Explicit realification

For concreteness, we list the real images of all 165 rays that have been previously enumerated and depicted in Fig. 2 of Ref. [7], as well as in Fig. 1 of Ref. [5], and Fig. 1 of Ref. [8]. Each row of Table I contains a label, the corresponding ray in $\mathbb{C}^3$, and its image in $\mathbb{R}^6$ under the canonical realification map (3) (with no additional phase factors). For a vector $(z_1,z_2,z_3) \in \mathbb{C}^3$ we write its real image as

$$(\Re z_1, \Re z_2, \Re z_3, \Im z_1, \Im z_2, \Im z_3) \in \mathbb{R}^6. \tag{14}$$

TABLE I: All 165 rays in $\mathbb{C}^3$ and their images in $\mathbb{R}^6$ under the canonical realification (3) (i.e. with all phase adjustments $\theta_k = 0$ as defined in (2)). This brings about spurious orthogonalities in $\mathbb{R}^6$, such as between u_1 and u_4 that are not present in $\mathbb{C}^3$. A strictly faithful embedding with no spurious orthogonalities requires the additional phase adjustments described in (15).

Label	$\mathbb{C}^3$ vector	$\mathbb{R}^6$ vector
a_{11}	$(1,0,0)$	$(1,0,0,\ 0,0,0)$
a_{21}	$(0,1,0)$	$(0,1,0,\ 0,0,0)$
a_{31}	$(0,0,1)$	$(0,0,1,\ 0,0,0)$
u_1	$(1,1,1)$	$(1,1,1,\ 0,0,0)$
u_2	$(1,\omega,\omega^2)$	$\left(1,-\frac{1}{2},-\frac{1}{2},\ 0,\frac{\sqrt{3}}{2},-\frac{\sqrt{3}}{2}\right)$
u_3	$(1,\omega^2,\omega)$	$\left(1,-\frac{1}{2},-\frac{1}{2},\ 0,-\frac{\sqrt{3}}{2},\frac{\sqrt{3}}{2}\right)$
u_4	$(1,\omega,\omega)$	$\left(1,-\frac{1}{2},-\frac{1}{2},\ 0,\frac{\sqrt{3}}{2},\frac{\sqrt{3}}{2}\right)$
u_5	$(1,\omega^2,1)$	$\left(1,-\frac{1}{2},1,\ 0,-\frac{\sqrt{3}}{2},0\right)$
u_6	$(1,1,\omega^2)$	$\left(1,1,-\frac{1}{2},\ 0,0,-\frac{\sqrt{3}}{2}\right)$
u_7	$(1,\omega^2,\omega^2)$	$\left(1,-\frac{1}{2},-\frac{1}{2},\ 0,-\frac{\sqrt{3}}{2},-\frac{\sqrt{3}}{2}\right)$
u_8	$(1,\omega,1)$	$\left(1,-\frac{1}{2},1,\ 0,\frac{\sqrt{3}}{2},0\right)$
u_9	$(1,1,\omega)$	$\left(1,1,-\frac{1}{2},\ 0,0,\frac{\sqrt{3}}{2}\right)$
b_{11}	$(0,1,1)$	$(0,1,1,\ 0,0,0)$
b_{12}	$(0,\omega,\omega^2)$	$\left(0,-\frac{1}{2},-\frac{1}{2},\ 0,\frac{\sqrt{3}}{2},-\frac{\sqrt{3}}{2}\right)$
b_{13}	$(0,\omega^2,\omega)$	$\left(0,-\frac{1}{2},-\frac{1}{2},\ 0,-\frac{\sqrt{3}}{2},\frac{\sqrt{3}}{2}\right)$
b_{21}	$(1,0,1)$	$(1,0,1,\ 0,0,0)$
b_{22}	$(1,0,\omega^2)$	$\left(1,0,-\frac{1}{2},\ 0,0,-\frac{\sqrt{3}}{2}\right)$
b_{23}	$(1,0,\omega)$	$\left(1,0,-\frac{1}{2},\ 0,0,\frac{\sqrt{3}}{2}\right)$
b_{31}	$(1,1,0)$	$(1,1,0,\ 0,0,0)$
b_{32}	$(1,\omega,0)$	$\left(1,-\frac{1}{2},0,\ 0,\frac{\sqrt{3}}{2},0\right)$
b_{33}	$(1,\omega^2,0)$	$\left(1,-\frac{1}{2},0,\ 0,-\frac{\sqrt{3}}{2},0\right)$
c_{11}	$(0,1,-1)$	$(0,1,-1,\ 0,0,0)$
c_{12}	$(0,\omega,-\omega^2)$	$\left(0,-\frac{1}{2},\frac{1}{2},\ 0,\frac{\sqrt{3}}{2},\frac{\sqrt{3}}{2}\right)$
c_{13}	$(0,\omega^2,-\omega)$	$\left(0,-\frac{1}{2},\frac{1}{2},\ 0,-\frac{\sqrt{3}}{2},-\frac{\sqrt{3}}{2}\right)$
c_{21}	$(-1,0,1)$	$(-1,0,1,\ 0,0,0)$
c_{22}	$(-\omega,0,1)$	$\left(\frac{1}{2},0,1,\ -\frac{\sqrt{3}}{2},0,0\right)$
c_{23}	$(-\omega^2,0,1)$	$\left(\frac{1}{2},0,1,\ \frac{\sqrt{3}}{2},0,0\right)$
c_{31}	$(1,-1,0)$	$(1,-1,0,\ 0,0,0)$
c_{32}	$(\omega^2,-1,0)$	$\left(-\frac{1}{2},-1,0,\ -\frac{\sqrt{3}}{2},0,0\right)$
c_{33}	$(\omega,-1,0)$	$\left(-\frac{1}{2},-1,0,\ \frac{\sqrt{3}}{2},0,0\right)$
d_{11}	$(-2,1,1)$	$(-2,1,1,\ 0,0,0)$
d_{12}	$(-2,\omega,\omega^2)$	$\left(-2,-\frac{1}{2},-\frac{1}{2},\ 0,\frac{\sqrt{3}}{2},-\frac{\sqrt{3}}{2}\right)$
d_{13}	$(-2,\omega^2,\omega)$	$\left(-2,-\frac{1}{2},-\frac{1}{2},\ 0,-\frac{\sqrt{3}}{2},\frac{\sqrt{3}}{2}\right)$
d_{14}	$(-2,\omega,\omega)$	$\left(-2,-\frac{1}{2},-\frac{1}{2},\ 0,\frac{\sqrt{3}}{2},\frac{\sqrt{3}}{2}\right)$
d_{15}	$(-2,\omega^2,1)$	$\left(-2,-\frac{1}{2},1,\ 0,-\frac{\sqrt{3}}{2},0\right)$
d_{16}	$(-2,1,\omega^2)$	$\left(-2,1,-\frac{1}{2},\ 0,0,-\frac{\sqrt{3}}{2}\right)$
d_{17}	$(-2,\omega^2,\omega^2)$	$\left(-2,-\frac{1}{2},-\frac{1}{2},\ 0,-\frac{\sqrt{3}}{2},-\frac{\sqrt{3}}{2}\right)$
d_{18}	$(-2,\omega,1)$	$\left(-2,-\frac{1}{2},1,\ 0,\frac{\sqrt{3}}{2},0\right)$
d_{19}	$(-2,1,\omega)$	$\left(-2,1,-\frac{1}{2},\ 0,0,\frac{\sqrt{3}}{2}\right)$
d_{21}	$(1,-2,1)$	$(1,-2,1,\ 0,0,0)$
d_{22}	$(\omega^2,-2,\omega)$	$\left(-\frac{1}{2},-2,-\frac{1}{2},\ -\frac{\sqrt{3}}{2},0,\frac{\sqrt{3}}{2}\right)$
d_{23}	$(\omega,-2,\omega^2)$	$\left(-\frac{1}{2},-2,-\frac{1}{2},\ \frac{\sqrt{3}}{2},0,-\frac{\sqrt{3}}{2}\right)$
d_{24}	$(\omega^2,-2,1)$	$\left(-\frac{1}{2},-2,1,\ -\frac{\sqrt{3}}{2},0,0\right)$
d_{25}	$(\omega,-2,\omega)$	$\left(-\frac{1}{2},-2,-\frac{1}{2},\ \frac{\sqrt{3}}{2},0,\frac{\sqrt{3}}{2}\right)$
d_{26}	$(1,-2,\omega^2)$	$\left(1,-2,-\frac{1}{2},\ 0,0,-\frac{\sqrt{3}}{2}\right)$
d_{27}	$(\omega,-2,1)$	$\left(-\frac{1}{2},-2,1,\ \frac{\sqrt{3}}{2},0,0\right)$
d_{28}	$(\omega^2,-2,\omega^2)$	$\left(-\frac{1}{2},-2,-\frac{1}{2},\ -\frac{\sqrt{3}}{2},0,-\frac{\sqrt{3}}{2}\right)$

Label	$\mathbb{C}^3$ vector	$\mathbb{R}^6$ vector
d_{29}	$(1,-2,\omega)$	$\left(1,-2,-\frac{1}{2},\ 0,0,\frac{\sqrt{3}}{2}\right)$
d_{31}	$(1,1,-2)$	$\left(1,1,-2,\ 0,0,0\right)$
d_{32}	$(\omega,\omega^2,-2)$	$\left(-\frac{1}{2},-\frac{1}{2},-2,\ \frac{\sqrt{3}}{2},-\frac{\sqrt{3}}{2},0\right)$
d_{33}	$(\omega^2,\omega,-2)$	$\left(-\frac{1}{2},-\frac{1}{2},-2,\ -\frac{\sqrt{3}}{2},\frac{\sqrt{3}}{2},0\right)$
d_{34}	$(\omega^2,1,-2)$	$\left(-\frac{1}{2},1,-2,\ -\frac{\sqrt{3}}{2},0,0\right)$
d_{35}	$(1,\omega^2,-2)$	$\left(1,-\frac{1}{2},-2,\ 0,-\frac{\sqrt{3}}{2},0\right)$
d_{36}	$(\omega,\omega,-2)$	$\left(-\frac{1}{2},-\frac{1}{2},-2,\ \frac{\sqrt{3}}{2},\frac{\sqrt{3}}{2},0\right)$
d_{37}	$(\omega,1,-2)$	$\left(-\frac{1}{2},1,-2,\ \frac{\sqrt{3}}{2},0,0\right)$
d_{38}	$(1,\omega,-2)$	$\left(1,-\frac{1}{2},-2,\ 0,\frac{\sqrt{3}}{2},0\right)$
d_{39}	$(\omega^2,\omega^2,-2)$	$\left(-\frac{1}{2},-\frac{1}{2},-2,\ -\frac{\sqrt{3}}{2},-\frac{\sqrt{3}}{2},0\right)$
b_{121}	$(1,1,-1)$	$\left(1,1,-1,\ 0,0,0\right)$
b_{122}	$(1,\omega,-\omega^2)$	$\left(1,-\frac{1}{2},\frac{1}{2},\ 0,\frac{\sqrt{3}}{2},\frac{\sqrt{3}}{2}\right)$
b_{123}	$(1,\omega^2,-\omega)$	$\left(1,-\frac{1}{2},\frac{1}{2},\ 0,-\frac{\sqrt{3}}{2},-\frac{\sqrt{3}}{2}\right)$
b_{124}	$(\omega,\omega^2,-\omega^2)$	$\left(-\frac{1}{2},-\frac{1}{2},\frac{1}{2},\ \frac{\sqrt{3}}{2},-\frac{\sqrt{3}}{2},\frac{\sqrt{3}}{2}\right)$
b_{125}	$(\omega,1,-\omega)$	$\left(-\frac{1}{2},1,\frac{1}{2},\ \frac{\sqrt{3}}{2},0,-\frac{\sqrt{3}}{2}\right)$
b_{126}	$(\omega,\omega,-1)$	$\left(-\frac{1}{2},-\frac{1}{2},-1,\ \frac{\sqrt{3}}{2},\frac{\sqrt{3}}{2},0\right)$
b_{127}	$(\omega^2,\omega,-\omega)$	$\left(-\frac{1}{2},-\frac{1}{2},\frac{1}{2},\ -\frac{\sqrt{3}}{2},\frac{\sqrt{3}}{2},-\frac{\sqrt{3}}{2}\right)$
b_{128}	$(\omega^2,1,-\omega^2)$	$\left(-\frac{1}{2},1,\frac{1}{2},\ -\frac{\sqrt{3}}{2},0,\frac{\sqrt{3}}{2}\right)$
b_{129}	$(\omega^2,\omega^2,-1)$	$\left(-\frac{1}{2},-\frac{1}{2},-1,\ -\frac{\sqrt{3}}{2},-\frac{\sqrt{3}}{2},0\right)$
b_{131}	$(1,-1,1)$	$\left(1,-1,1,\ 0,0,0\right)$
b_{132}	$(1,-\omega,\omega^2)$	$\left(1,\frac{1}{2},-\frac{1}{2},\ 0,-\frac{\sqrt{3}}{2},-\frac{\sqrt{3}}{2}\right)$
b_{133}	$(1,-\omega^2,\omega)$	$\left(1,\frac{1}{2},-\frac{1}{2},\ 0,\frac{\sqrt{3}}{2},\frac{\sqrt{3}}{2}\right)$
b_{134}	$(\omega,-\omega^2,\omega^2)$	$\left(-\frac{1}{2},\frac{1}{2},-\frac{1}{2},\ \frac{\sqrt{3}}{2},\frac{\sqrt{3}}{2},-\frac{\sqrt{3}}{2}\right)$
b_{135}	$(\omega,-1,\omega)$	$\left(-\frac{1}{2},-1,-\frac{1}{2},\ \frac{\sqrt{3}}{2},0,\frac{\sqrt{3}}{2}\right)$
b_{136}	$(\omega,-\omega,1)$	$\left(-\frac{1}{2},\frac{1}{2},1,\ \frac{\sqrt{3}}{2},-\frac{\sqrt{3}}{2},0\right)$
b_{137}	$(\omega^2,-\omega,\omega)$	$\left(-\frac{1}{2},\frac{1}{2},-\frac{1}{2},\ -\frac{\sqrt{3}}{2},-\frac{\sqrt{3}}{2},\frac{\sqrt{3}}{2}\right)$
b_{138}	$(\omega^2,-1,\omega^2)$	$\left(-\frac{1}{2},-1,-\frac{1}{2},\ -\frac{\sqrt{3}}{2},0,-\frac{\sqrt{3}}{2}\right)$
b_{139}	$(\omega^2,-\omega^2,1)$	$\left(-\frac{1}{2},\frac{1}{2},1,\ -\frac{\sqrt{3}}{2},\frac{\sqrt{3}}{2},0\right)$
b_{231}	$(-1,1,1)$	$\left(-1,1,1,\ 0,0,0\right)$
b_{232}	$(-1,\omega,\omega^2)$	$\left(-1,-\frac{1}{2},-\frac{1}{2},\ 0,\frac{\sqrt{3}}{2},-\frac{\sqrt{3}}{2}\right)$
b_{233}	$(-1,\omega^2,\omega)$	$\left(-1,-\frac{1}{2},-\frac{1}{2},\ 0,-\frac{\sqrt{3}}{2},\frac{\sqrt{3}}{2}\right)$
b_{234}	$(-\omega,\omega^2,\omega^2)$	$\left(\frac{1}{2},-\frac{1}{2},-\frac{1}{2},\ -\frac{\sqrt{3}}{2},-\frac{\sqrt{3}}{2},-\frac{\sqrt{3}}{2}\right)$
b_{235}	$(-\omega,1,\omega)$	$\left(\frac{1}{2},1,-\frac{1}{2},\ -\frac{\sqrt{3}}{2},0,\frac{\sqrt{3}}{2}\right)$
b_{236}	$(-\omega,\omega,1)$	$\left(\frac{1}{2},-\frac{1}{2},1,\ -\frac{\sqrt{3}}{2},\frac{\sqrt{3}}{2},0\right)$
b_{237}	$(-\omega^2,\omega,\omega)$	$\left(\frac{1}{2},-\frac{1}{2},-\frac{1}{2},\ \frac{\sqrt{3}}{2},\frac{\sqrt{3}}{2},\frac{\sqrt{3}}{2}\right)$
b_{238}	$(-\omega^2,1,\omega^2)$	$\left(\frac{1}{2},1,-\frac{1}{2},\ \frac{\sqrt{3}}{2},0,-\frac{\sqrt{3}}{2}\right)$
b_{239}	$(-\omega^2,\omega^2,1)$	$\left(\frac{1}{2},-\frac{1}{2},1,\ \frac{\sqrt{3}}{2},-\frac{\sqrt{3}}{2},0\right)$
e_{11}	$(1,2,1)$	$\left(1,2,1,\ 0,0,0\right)$
e_{12}	$(\omega^2,2,\omega)$	$\left(-\frac{1}{2},2,-\frac{1}{2},\ -\frac{\sqrt{3}}{2},0,\frac{\sqrt{3}}{2}\right)$

Label	$\mathbb{C}^3$ vector	$\mathbb{R}^6$ vector
e_{13}	$(\omega,2,\omega^2)$	$\left(-\frac{1}{2},2,-\frac{1}{2},\ \frac{\sqrt{3}}{2},0,-\frac{\sqrt{3}}{2}\right)$
e_{14}	$(\omega,2\omega^2,\omega^2)$	$\left(-\frac{1}{2},-1,-\frac{1}{2},\ \frac{\sqrt{3}}{2},-\sqrt{3},-\frac{\sqrt{3}}{2}\right)$
e_{15}	$(1,2\omega^2,1)$	$\left(1,-1,1,\ 0,-\sqrt{3},0\right)$
e_{16}	$(\omega^2,2\omega^2,\omega)$	$\left(-\frac{1}{2},-1,-\frac{1}{2},\ -\frac{\sqrt{3}}{2},-\sqrt{3},\frac{\sqrt{3}}{2}\right)$
e_{17}	$(\omega^2,2\omega,\omega)$	$\left(-\frac{1}{2},-1,-\frac{1}{2},\ -\frac{\sqrt{3}}{2},\sqrt{3},\frac{\sqrt{3}}{2}\right)$
e_{18}	$(1,2\omega,1)$	$\left(1,-1,1,\ 0,\sqrt{3},0\right)$
e_{19}	$(\omega,2\omega,\omega^2)$	$\left(-\frac{1}{2},-1,-\frac{1}{2},\ \frac{\sqrt{3}}{2},\sqrt{3},-\frac{\sqrt{3}}{2}\right)$
e_{21}	$(1,1,2)$	$\left(1,1,2,\ 0,0,0\right)$
e_{22}	$(\omega,\omega^2,2)$	$\left(-\frac{1}{2},-\frac{1}{2},2,\ \frac{\sqrt{3}}{2},-\frac{\sqrt{3}}{2},0\right)$
e_{23}	$(\omega^2,\omega,2)$	$\left(-\frac{1}{2},-\frac{1}{2},2,\ -\frac{\sqrt{3}}{2},\frac{\sqrt{3}}{2},0\right)$
e_{24}	$(\omega,\omega^2,2\omega^2)$	$\left(-\frac{1}{2},-\frac{1}{2},-1,\ \frac{\sqrt{3}}{2},-\frac{\sqrt{3}}{2},-\sqrt{3}\right)$
e_{25}	$(\omega^2,\omega,2\omega^2)$	$\left(-\frac{1}{2},-\frac{1}{2},-1,\ -\frac{\sqrt{3}}{2},\frac{\sqrt{3}}{2},-\sqrt{3}\right)$
e_{26}	$(1,1,2\omega^2)$	$\left(1,1,-1,\ 0,0,-\sqrt{3}\right)$
e_{27}	$(\omega^2,\omega,2\omega)$	$\left(-\frac{1}{2},-\frac{1}{2},-1,\ -\frac{\sqrt{3}}{2},\frac{\sqrt{3}}{2},\sqrt{3}\right)$
e_{28}	$(\omega,\omega^2,2\omega)$	$\left(-\frac{1}{2},-\frac{1}{2},-1,\ \frac{\sqrt{3}}{2},-\frac{\sqrt{3}}{2},\sqrt{3}\right)$
e_{29}	$(1,1,2\omega)$	$\left(1,1,-1,\ 0,0,\sqrt{3}\right)$
e_{31}	$(2,1,1)$	$\left(2,1,1,\ 0,0,0\right)$
e_{32}	$(2,\omega,\omega^2)$	$\left(2,-\frac{1}{2},-\frac{1}{2},\ 0,\frac{\sqrt{3}}{2},-\frac{\sqrt{3}}{2}\right)$
e_{33}	$(2,\omega^2,\omega)$	$\left(2,-\frac{1}{2},-\frac{1}{2},\ 0,-\frac{\sqrt{3}}{2},\frac{\sqrt{3}}{2}\right)$
e_{34}	$(2\omega^2,1,1)$	$\left(-1,1,1,\ -\sqrt{3},0,0\right)$
e_{35}	$(2\omega^2,\omega,\omega^2)$	$\left(-1,-\frac{1}{2},-\frac{1}{2},\ -\sqrt{3},\frac{\sqrt{3}}{2},-\frac{\sqrt{3}}{2}\right)$
e_{36}	$(2\omega^2,\omega^2,\omega)$	$\left(-1,-\frac{1}{2},-\frac{1}{2},\ -\sqrt{3},-\frac{\sqrt{3}}{2},\frac{\sqrt{3}}{2}\right)$
e_{37}	$(2\omega,1,1)$	$\left(-1,1,1,\ \sqrt{3},0,0\right)$
e_{38}	$(2\omega,\omega^2,\omega)$	$\left(-1,-\frac{1}{2},-\frac{1}{2},\ \sqrt{3},-\frac{\sqrt{3}}{2},\frac{\sqrt{3}}{2}\right)$
e_{39}	$(2\omega,\omega,\omega^2)$	$\left(-1,-\frac{1}{2},-\frac{1}{2},\ \sqrt{3},\frac{\sqrt{3}}{2},-\frac{\sqrt{3}}{2}\right)$
b_{1121}	$(2,-1,1)$	$\left(2,-1,1,\ 0,0,0\right)$
b_{1122}	$(2,-\omega,\omega^2)$	$\left(2,\frac{1}{2},-\frac{1}{2},\ 0,-\frac{\sqrt{3}}{2},-\frac{\sqrt{3}}{2}\right)$
b_{1123}	$(2,-\omega^2,\omega)$	$\left(2,\frac{1}{2},-\frac{1}{2},\ 0,\frac{\sqrt{3}}{2},\frac{\sqrt{3}}{2}\right)$
b_{1124}	$(2,-\omega,\omega)$	$\left(2,\frac{1}{2},-\frac{1}{2},\ 0,-\frac{\sqrt{3}}{2},\frac{\sqrt{3}}{2}\right)$
b_{1125}	$(2,-\omega^2,1)$	$\left(2,\frac{1}{2},1,\ 0,\frac{\sqrt{3}}{2},0\right)$
b_{1126}	$(2,-1,\omega^2)$	$\left(2,-1,-\frac{1}{2},\ 0,0,-\frac{\sqrt{3}}{2}\right)$
b_{1127}	$(2,-\omega^2,\omega^2)$	$\left(2,\frac{1}{2},-\frac{1}{2},\ 0,\frac{\sqrt{3}}{2},-\frac{\sqrt{3}}{2}\right)$
b_{1128}	$(2,-\omega,1)$	$\left(2,\frac{1}{2},1,\ 0,-\frac{\sqrt{3}}{2},0\right)$
b_{1129}	$(2,-1,\omega)$	$\left(2,-1,-\frac{1}{2},\ 0,0,\frac{\sqrt{3}}{2}\right)$
b_{2121}	$(-1,2,1)$	$\left(-1,2,1,\ 0,0,0\right)$
b_{2122}	$(-1,2\omega,\omega^2)$	$\left(-1,-1,-\frac{1}{2},\ 0,\sqrt{3},-\frac{\sqrt{3}}{2}\right)$
b_{2123}	$(-1,2\omega^2,\omega)$	$\left(-1,-1,-\frac{1}{2},\ 0,-\sqrt{3},\frac{\sqrt{3}}{2}\right)$
b_{2124}	$(-1,2\omega,\omega)$	$\left(-1,-1,-\frac{1}{2},\ 0,\sqrt{3},\frac{\sqrt{3}}{2}\right)$
b_{2125}	$(-1,2\omega^2,1)$	$\left(-1,-1,1,\ 0,-\sqrt{3},0\right)$

Label	$\mathbb{C}^3$ vector	$\mathbb{R}^6$ vector
b_{2126}	$(-1, 2, \omega^2)$	$\left(-1, 2, -\frac{1}{2}, 0, 0, -\frac{\sqrt{3}}{2}\right)$
b_{2127}	$(-1, 2\omega^2, \omega^2)$	$\left(-1, -1, -\frac{1}{2}, 0, -\sqrt{3}, -\frac{\sqrt{3}}{2}\right)$
b_{2128}	$(-1, 2\omega, 1)$	$\left(-1, -1, 1, 0, \sqrt{3}, 0\right)$
b_{2129}	$(-1, 2, \omega)$	$\left(-1, 2, -\frac{1}{2}, 0, 0, \frac{\sqrt{3}}{2}\right)$
b_{1131}	$(2, 1, -1)$	$\left(2, 1, -1, 0, 0, 0\right)$
b_{1132}	$(2, \omega, -\omega^2)$	$\left(2, -\frac{1}{2}, \frac{1}{2}, 0, \frac{\sqrt{3}}{2}, \frac{\sqrt{3}}{2}\right)$
b_{1133}	$(2, \omega^2, -\omega)$	$\left(2, -\frac{1}{2}, \frac{1}{2}, 0, -\frac{\sqrt{3}}{2}, -\frac{\sqrt{3}}{2}\right)$
b_{1134}	$(2, \omega, -\omega)$	$\left(2, -\frac{1}{2}, \frac{1}{2}, 0, \frac{\sqrt{3}}{2}, -\frac{\sqrt{3}}{2}\right)$
b_{1135}	$(2, \omega^2, -1)$	$\left(2, -\frac{1}{2}, -1, 0, -\frac{\sqrt{3}}{2}, 0\right)$
b_{1136}	$(2, 1, -\omega^2)$	$\left(2, 1, \frac{1}{2}, 0, 0, \frac{\sqrt{3}}{2}\right)$
b_{1137}	$(2, \omega^2, -\omega^2)$	$\left(2, -\frac{1}{2}, \frac{1}{2}, 0, -\frac{\sqrt{3}}{2}, \frac{\sqrt{3}}{2}\right)$
b_{1138}	$(2, \omega, -1)$	$\left(2, -\frac{1}{2}, -1, 0, \frac{\sqrt{3}}{2}, 0\right)$
b_{1139}	$(2, 1, -\omega)$	$\left(2, 1, \frac{1}{2}, 0, 0, -\frac{\sqrt{3}}{2}\right)$
b_{3131}	$(-1, 1, 2)$	$\left(-1, 1, 2, 0, 0, 0\right)$
b_{3132}	$(-1, \omega, 2\omega^2)$	$\left(-1, -\frac{1}{2}, -1, 0, \frac{\sqrt{3}}{2}, -\sqrt{3}\right)$
b_{3133}	$(-1, \omega^2, 2\omega)$	$\left(-1, -\frac{1}{2}, -1, 0, -\frac{\sqrt{3}}{2}, \sqrt{3}\right)$
b_{3134}	$(-1, \omega, 2\omega)$	$\left(-1, -\frac{1}{2}, -1, 0, \frac{\sqrt{3}}{2}, \sqrt{3}\right)$
b_{3135}	$(-1, \omega^2, 2)$	$\left(-1, -\frac{1}{2}, 2, 0, -\frac{\sqrt{3}}{2}, 0\right)$
b_{3136}	$(-1, 1, 2\omega^2)$	$\left(-1, 1, -1, 0, 0, -\sqrt{3}\right)$
b_{3137}	$(-1, \omega^2, 2\omega^2)$	$\left(-1, -\frac{1}{2}, -1, 0, -\frac{\sqrt{3}}{2}, -\sqrt{3}\right)$
b_{3138}	$(-1, \omega, 2)$	$\left(-1, -\frac{1}{2}, 2, 0, \frac{\sqrt{3}}{2}, 0\right)$
b_{3139}	$(-1, 1, 2\omega)$	$\left(-1, 1, -1, 0, 0, \sqrt{3}\right)$
b_{2231}	$(1, 2, -1)$	$\left(1, 2, -1, 0, 0, 0\right)$
b_{2232}	$(1, 2\omega, -\omega^2)$	$\left(1, -1, \frac{1}{2}, 0, \sqrt{3}, \frac{\sqrt{3}}{2}\right)$
b_{2233}	$(1, 2\omega^2, -\omega)$	$\left(1, -1, \frac{1}{2}, 0, -\sqrt{3}, -\frac{\sqrt{3}}{2}\right)$
b_{2234}	$(1, 2\omega, -\omega)$	$\left(1, -1, \frac{1}{2}, 0, \sqrt{3}, -\frac{\sqrt{3}}{2}\right)$
b_{2235}	$(1, 2\omega^2, -1)$	$\left(1, -1, -1, 0, -\sqrt{3}, 0\right)$
b_{2236}	$(1, 2, -\omega^2)$	$\left(1, 2, \frac{1}{2}, 0, 0, \frac{\sqrt{3}}{2}\right)$
b_{2237}	$(1, 2\omega^2, -\omega^2)$	$\left(1, -1, \frac{1}{2}, 0, -\sqrt{3}, \frac{\sqrt{3}}{2}\right)$
b_{2238}	$(1, 2\omega, -1)$	$\left(1, -1, -1, 0, \sqrt{3}, 0\right)$
b_{2239}	$(1, 2, -\omega)$	$\left(1, 2, \frac{1}{2}, 0, 0, -\frac{\sqrt{3}}{2}\right)$
b_{3231}	$(1, -1, 2)$	$\left(1, -1, 2, 0, 0, 0\right)$
b_{3232}	$(1, -\omega, 2\omega^2)$	$\left(1, \frac{1}{2}, -1, 0, -\frac{\sqrt{3}}{2}, -\sqrt{3}\right)$
b_{3233}	$(1, -\omega^2, 2\omega)$	$\left(1, \frac{1}{2}, -1, 0, \frac{\sqrt{3}}{2}, \sqrt{3}\right)$
b_{3234}	$(1, -\omega, 2\omega)$	$\left(1, \frac{1}{2}, -1, 0, -\frac{\sqrt{3}}{2}, \sqrt{3}\right)$
b_{3235}	$(1, -\omega^2, 2)$	$\left(1, \frac{1}{2}, 2, 0, \frac{\sqrt{3}}{2}, 0\right)$
b_{3236}	$(1, -1, 2\omega^2)$	$\left(1, -1, -1, 0, 0, -\sqrt{3}\right)$
b_{3237}	$(1, -\omega^2, 2\omega^2)$	$\left(1, \frac{1}{2}, -1, 0, \frac{\sqrt{3}}{2}, -\sqrt{3}\right)$
b_{3238}	$(1, -\omega, 2)$	$\left(1, \frac{1}{2}, 2, 0, -\frac{\sqrt{3}}{2}, 0\right)$
b_{3239}	$(1, -1, 2\omega)$	$\left(1, -1, -1, 0, 0, \sqrt{3}\right)$

The 165 rays listed in Table I are defined only up to overall complex phases. In order to compare their realifications in $\mathbb{R}^6$, it is necessary to make a consistent phase choice. A direct symbolic computation shows that one can assign to each ray $v_k \in \mathbb{C}^3$ a phase of the form

$$\theta_k = \frac{n_k \pi}{K}, \qquad n_k \in \mathbb{Z}, \tag{15}$$

where $K = 1009$ stands for some nonunique large integer (we choose a large prime number), such that the following two conditions hold simultaneously: (i) whenever $\langle v_i, v_j \rangle = 0$ in $\mathbb{C}^3$, the corresponding realified vectors $w_i, w_j \in \mathbb{R}^6$ satisfy $w_i \cdot w_j = 0$; and (ii) whenever $\langle v_i, v_j \rangle \neq 0$, the realifications satisfy $w_i \cdot w_j \neq 0$. In particular, this phase choice introduces no spurious orthogonality relations in $\mathbb{R}^6$ and preserves all of the orthogonality relations that define the original KS hypergraph in $\mathbb{C}^3$. We verified the existence of such phases numerically with a simple backtracking script and confirmed that $K = 1009$ suffices. The existence of such a rational phase assignment demonstrates that the entire configuration admits an analytic (real-algebraic) realification without altering its orthogonality structure.

IV. BREAKDOWN OF ADMISSIBILITY CRITERIA IN $\mathbb{R}^6$

The Kochen–Specker contradiction arises from the impossibility of assigning a two-valued state $v(P) \in \{0, 1\}$ to every projector P in a set $\mathcal{S}$ such that the assignment satisfies two fundamental admissibility criteria:

1. **Exclusivity:** For any pair of mutually orthogonal projectors P_i, P_j, the values must satisfy $v(P_i + P_j) = v(P_i) + v(P_j)$. In particular, if $v(P_i) = 1$, then $v(P_j) = 0$ for all $P_j \perp P_i$.

2. **Completeness:** The assignment must respect the identity observable, $v(\mathbb{I}) = 1$.

In the original complex space $\mathbb{C}^3$, any context $\mathcal{C} = \{P_1, P_2, P_3\}$ consisting of rank-1 projectors onto an orthonormal basis satisfies $\sum_{i=1}^{3} P_i = \mathbb{I}_{\mathbb{C}^3}$. Combining Exclusivity and Completeness yields the strict sum rule:

$$\sum_{i=1}^{3} v(P_i) = v\left(\sum_{i=1}^{3} P_i\right) = v(\mathbb{I}_{\mathbb{C}^3}) = 1. \tag{16}$$

Thus, within every context in $\mathbb{C}^3$, exactly one projector must be assigned the value 1.

However, under the faithful embedding into $\mathbb{R}^6$, we denote the projectors mapped from a single complex context by $\{\Pi_1, \Pi_2, \Pi_3\}$; more generally, we write $\Pi_{i,j}$ for the projector onto the i-th ray in the j-th context. The triple $\{\Pi_1, \Pi_2, \Pi_3\}$ spans only a three-dimensional subspace. While they remain mutually orthogonal (satisfying Exclusivity), they do not sum to the identity of the embedding space:

$$\sum_{i=1}^{3} \Pi_i = \Pi_{\text{sub}} \neq \mathbb{I}_{\mathbb{R}^6}. \tag{17}$$

Completeness applies only to the full space $\mathbb{R}^6$. To invoke it, one must extend the set $\{\Pi_1, \Pi_2, \Pi_3\}$ with orthogonal projectors $\{\Pi_4, \Pi_5, \Pi_6\}$ completing the basis, such that $\sum_{k=1}^{6} \Pi_k = \mathbb{I}_{\mathbb{R}^6}$. The Completeness criterion then imposes:

$$\sum_{k=1}^{6} v(\Pi_k) = 1. \tag{18}$$

Restricted to the original embedded context, the strict equality of Eq. (16) relaxes to an inequality:

$$0 \leq \sum_{i=1}^{3} v(\Pi_i) \leq 1. \tag{19}$$

This relaxation permits the assignment $v(\Pi_1) = v(\Pi_2) = v(\Pi_3) = 0$ (provided $v(\Pi_k) = 1$ for some $k \in \{4, 5, 6\}$). Consequently, one can assign the value 0 to all 165 embedded rays in $\mathbb{R}^6$ simultaneously without violating Exclusivity or Completeness, thereby avoiding the Kochen–Specker contradiction. In $\mathbb{C}^3$ the same "all-zero" assignment on the 165 rays would yield $\sum_{i=1}^{3} v(\Pi_i) = 0$ in *every* context, thereby violating the completeness requirement $v(\mathbb{I}_{\mathbb{C}^3}) = 1$.

While Eq. (19) allows for the local possibility that a specific embedded context sums to 1 (i.e., retains the "1" within the embedded subspace), the Kochen–Specker obstruction in $\mathbb{C}^3$ manifests globally in $\mathbb{R}^6$ as a strict bound on the total valuation. Considering the complete set of $|\mathcal{C}| = 130$ contexts in the configuration, as analysed in detail by Navara and Svozil [8], if it were possible to assign values such that every embedded context retained the value 1, the total sum would be exactly 130. However, since this is equivalent to a Kochen–Specker coloring in $\mathbb{C}^3$ (which is impossible), the global sum is strictly bounded:

$$0 \leq \sum_{j=1}^{130} \sum_{i=1}^{3} v(\Pi_{i,j}) < 130. \tag{20}$$

The lower bound of 0 corresponds to the "all-zeros" assignment mentioned earlier where the value 1 is placed in the orthogonal complement for every context. The strict upper bound (< 130) implies that there is a minimum non-zero deficit. Numerical optimization over all two-valued assignments compatible with Exclusivity and Completeness confirms that at least two contexts must unavoidably surrender the value 1 to the extra dimensions of $\mathbb{R}^6$ to resolve the logical contradiction. Thus, the final bound is tightened to:

$$0 \leq \sum_{j=1}^{130} \sum_{i=1}^{3} v(\Pi_{i,j}) \leq 128. \tag{21}$$

V. DISCUSSION

We have described a simple phase-adjusted realification procedure that embeds any finite set of rays in $\mathbb{C}^3$ into $\mathbb{R}^6$ in a way that preserves and reflects complex orthogonality. The construction is purely algebraic and uses only the phase freedom inherent in projective rays. Applied to the 165-ray configuration built from MUBs, it yields a faithful orthogonal representation in $\mathbb{R}^6$ (after suitable phase choices), and, by trivial extension, in $\mathbb{R}^7$ and above.

Table I lists all 165 rays explicitly in $\mathbb{C}^3$ together with their canonical realifications in $\mathbb{R}^6$. For applications requiring strict faithfulness (no extra orthogonality among non-orthogonal rays), one additionally picks phases θ_k as in Sec. II before applying Φ_0, which simply rotates each pair $(\Re z_j, \Im z_j)$ in its two-dimensional real subspace.

Whereas, in its original complex realisation in $\mathbb{C}^3$, the 165-ray, 130-context configuration is Kochen–Specker uncolourable (there is no two-valued state assigning 0 and 1 such that in each 3-element context exactly one projector has value 1 [5]), the same set of rays, when embedded in $\mathbb{R}^6$, does admit classical two-valued states if we keep only the embedded 165 rays and the original 3-element contexts.

The reason is that, in $\mathbb{R}^6$, each such triple now spans only a 3-dimensional subspace and is no longer a maximal context: it can always be completed to a full orthonormal basis of $\mathbb{R}^6$ by adding three extra vectors not belonging to our configuration. A valuation that assigns the value 0 to all 165 given rays, and places the value 1 on one of these additional basis vectors for each completed context, then satisfies all Kochen–Specker constraints on the restricted set. Because there is a continuum of choices for the orthonormal complements (each 3-dimensional subspace has infinitely many orthonormal bases of its orthogonal complement), one can arrange the completions so that no consistency problems arise across different contexts.

This contrast highlights that a faithful orthogonal representation of a KS hypergraph in a higher-dimensional real space does not automatically preserve KS uncolourability: classical two-valued interpretations exist for the embedded configuration in $\mathbb{R}^6$, whereas they are forbidden for its realisation as maximal contexts in $\mathbb{C}^3$.

There are two conceptual messages that emerge from this analysis. First, our construction recovers and reformulates, in explicitly MUB-based terms, the central conclusion of Harding and Salinas Schmeis [7]: in three dimensions there exist finite orthogonality configurations of rank-1 projectors that are representable in $\mathbb{C}^3$ but admit no faithful orthogonal representation [15] in $\mathbb{R}^3$. In particular, the triple Yu–Oh construction [5, 6], which can be implemented experimentally for a single qutrit (for instance via generalized multiport interferometers [16]), realises such a configuration: its hypergraph of orthogonality relations can be realised by unit vectors in $\mathbb{C}^3$ but not by unit vectors in $\mathbb{R}^3$.

Second, the same triple Yu–Oh configuration, when completed to a Kochen–Specker set in $\mathbb{C}^3$, is KS-uncolourable: there is no two-valued state (global 0–1 assignment) compatible with all its 3-element contexts. Our phase-adjusted embedding shows that this configuration does admit a faithful orthogonal representation in $\mathbb{R}^6$ (and trivially in $\mathbb{R}^D$ for all $D \geq 6$). However, in $\mathbb{R}^6$ each 3-element context spans only a 3-dimensional subspace and is no longer maximal; it can always be extended to a full orthonormal basis of $\mathbb{R}^6$ by adding three extra vectors outside the given set. One can then construct classical two-valued states by assigning the value 0 to all 165 embedded rays and the value 1 to one of the additional basis vectors in each completed context. Because each 3-dimensional subspace has a continuum of orthogonal complements, the completions can be chosen consistently across contexts. Thus, the complex realisation in $\mathbb{C}^3$ is inherently KS-nonclassical, whereas the real realisation in $\mathbb{R}^6$ admits classical hidden-variable (two-valued state) models once the contexts are interpreted as maximal in the larger space.

This sharpens and complements earlier tests that distinguish real and complex quantum theory at the level of measurement statistics [2–4]: here the separation appears already at the level of the logical structure, in the existence or absence of two-valued states on the same finite orthogonality hypergraph when realised in $\mathbb{C}^3$ versus in $\mathbb{R}^6$.

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